

Module 2 Task: Endpoint Defense & Memory Analysis

Role: Senior SOC Tier 2 Analyst

Objective: Analyze forensic artifacts to identify an active compromise.

1. Incident Overview

A workstation in the Finance department triggered a generic Suspicious Network Connection alert. The local antivirus did not detect any malicious files on the disk. A memory dump was captured, and the following Volatility reports have been provided for your analysis:

1. pstree_output.txt (Process Tree Analysis)
2. malfind_results.txt (Memory Injection/In-memory artifacts)
3. netscan_results.txt (Network Connection artifacts)

2. Your Mission

You must analyze the provided raw data to reconstruct the attack. You are expected to find the Signal in the Noise. Several legitimate system processes are running. Your job is to identify which one is not what it seems.

Recommended Analysis Tools

To perform this analysis efficiently, you should use tools that allow for deep searching and side-by-side comparison of data:

- **Advanced Text Editors:** **VS Code** (use Split View to compare pstree and netscan side-by-side) or **Notepad++**.
- **Command Line Utilities:** Use **grep** (Linux/Mac) or **findstr** (Windows) to quickly filter PIDs across all three files.
- **Hex/Text Viewers:** Use **CyberChef** if you need to decode any strings found in the malfind output.
- **Knowledge Bases:** Refer to the **MITRE ATT&CK Framework** and **LOLBAS (Living Off The Land Binaries and Scripts)** to verify if a process's behavior matches known attack patterns.

Required Deliverables

In your final investigation report, you must answer the following:

1. **Identify the Compromised Process:** Provide the Name and PID of the process acting as the vessel for the malware.
2. **Parent-Child Analysis:** Identify the Parent Process (Name and PID) of the malicious entry. Explain why this relationship is architecturally significant or suspicious in a Windows environment.

3. **Memory Evidence:** Cite the specific memory protection flags and Magic Bytes found in the malfind report that prove code injection has occurred.
4. **C2 Identification:** Provide the Destination IP address and Port the compromised process is communicating with.
5. **Attack Classification:** Based on the evidence (Process Tree + Memory Injection + Network Call), what specific type of memory-resident attack is this? (e.g., Process Hollowing, DLL Injection, or Reflective Loading).

3. Evidence Guidelines

- **Do not** assume a process is safe just because it has a standard Windows name.
- Compare the netscan results with the pstree results to see which processes are actually talking to the internet.
- Look for inconsistencies between how Windows *should* work and what the data shows.

Submission Instructions: Your report should be professional and suitable for a SOC Manager. Use technical terminology (e.g., Parent-Child relationship, Memory Protection, Hexadecimal Headers).